

Recovery-Linked Quotient Observability: A Partial-Identification Framework for Report-Hidden Content Dynamics

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Abstract

A neural response observed before behavioral recovery does not by itself show that a later behaviorally effective content distinction was already present. The response may instead reflect weak but nonzero behavior, arousal, stimulus processing, changing measurement dynamics, nonstationary mixtures, or a mode created only after recovery. This paper develops **Recovery-Linked Quotient Observability** (RLQO), a model-relative framework for separating these possibilities.

For a prespecified scalar intervention contrast c , let $\mathcal{H}_c^J(\rho)$ and $\mathcal{H}_c^B(\rho)$ be block-Hankel matrices of dynamic interventional kernels from that contrast to, respectively, joint neural-behavioral and selected behavioral outputs at recovery stage ρ . The relative rank

$$q_c(\rho) = \text{rank } \mathcal{H}_c^J(\rho) - \text{rank } \mathcal{H}_c^B(\rho)$$

equals

$$\dim(\mathcal{U}_B(\rho)/\mathcal{U}_J(\rho))$$

once the chosen horizons saturate content reachability and observability. Here $\mathcal{U}_B$ and $\mathcal{U}_J$ are the content-reachable subspaces unobservable from behavioral and joint outputs. Under reproducible point-state preparation, an identifiable state-independent translation-noise family, common future probes, and an attainable-set condition, q_c is also the dimension of the kernel of the canonical linear map from joint to behavioral predictive-state quotients. Without those additional assumptions, it is only an algebraic relative-observability invariant.

A continuous transport theorem is retained only as an oracle-level structural result: within a fixed-degree analytic minimal transfer family with isolated simple poles, pole branches and residues are invariant under latent-state similarity. It does not identify continuous branches from finitely sampled stages. For finite

longitudinal data, the paper instead defines lower and upper transport indices, Ω_c^- and Ω_c^+ , by optimizing over all branch matchings and Lipschitz interpolants consistent with simultaneous confidence sets, a fixed McMillan degree, prespecified measurement metrics, and independently justified drift bounds. If $\Omega_c^- > 0$, every admissible transport history contains a branch that remains sensitive to the same imposed contrast, is neurally present throughout the interval, lies below an early behavioral-silence margin, and exceeds a later behavioral-opening margin. If $\Omega_c^- = 0 < \Omega_c^+$, transport is only partially identified.

A finite passive record remains insufficient in a broad model class that permits endogenous assignment and unbounded delay order. Conversely, a stipulated non-conscious cache can satisfy both the relative-rank and transport criteria. RLQO therefore concerns selected-input, selected-output controlled dynamics. It neither identifies biological-mode identity outside its model class nor establishes phenomenal consciousness.

Research Question and Hypothesis

Research question

Under what conditions can longitudinal intervention data support the restricted claim that:

a dynamical mode sensitive to a prespecified experimental contrast was neurally measurable before it influenced a selected behavioral output, and a compatible transfer-function branch later became behaviorally effective?

This question is narrower than whether a patient was aware. It is also narrower than whether an early neural response predicts recovery. Prediction can arise from injury severity, arousal, medication, sensor quality, spontaneous state mixtures, or general physiological integrity without identifying a report-hidden content-responsive mode.

The alternatives of interest are:

1. a weak behavioral response was already present but fell below detection;
2. neural and behavioral responses share a changing arousal or measurement gain;
3. a motor or communicative pathway recovers without an independently content-reachable early neural distinction;
4. a late mode is created, replaced, or switched rather than transported;
5. a pole persists while its loading rotates to a different input contrast;
6. changing hemodynamic, sensor, reference, or preprocessing dynamics create apparent transport;
7. an early association is generated by a fixed passive policy rather than an identified intervention; or
8. aggregation across patients or states creates modes not present in any individual system.

Operational target

Let c denote one prespecified scalar contrast generated from a randomized or otherwise causally identified intervention. Let

$$\rho_0 < \rho_1 < \cdots < \rho_K$$

be within-person stages scheduled independently of the behavioral opening under study. Stages may be calendar times or preregistered clinical epochs, but they must not be selected retrospectively as “the last unresponsive session” and “the first responsive session.”

An exact, model-relative recovery-linked quotient-observability relation requires:

1. **Contrast reachability:** the imposed contrast reaches a nonzero latent direction at ρ_0 .
2. **Early neural observability:** the direction changes future neural-output laws.
3. **Early behavioral unobservability:** it does not change future laws of the selected behavioral output.
4. **Attainability:** the relevant predictive distinction can be generated by two reproducible admissible preparations, not merely by taking a formal linear span.
5. **Branch continuity:** within a declared fixed-order transport model, the same transfer-function branch remains isolated and sensitive to the same imposed contrast.
6. **Continuous neural presence:** the branch’s neural residue remains nonzero throughout the modeled interval.
7. **Later behavioral opening:** the branch acquires a nonzero behavioral residue at the later stage.

The empirical version replaces exact zeros and nonzeros with preregistered equivalence and detection margins. It also replaces unique analytic continuation with lower and upper indices over all admissible branch histories.

“Content” is operational. It denotes the imposed contrast, not necessarily a comprehended instruction, intended imagery, semantic representation, conscious content, or subject-owned experience. A command cue is an intervention on a physical cue unless mediation through comprehension and task performance is independently supported.

Structural hypothesis

For an exact finite-dimensional linear system with sufficient Hankel horizons,

$$q_c(\rho_0) > 0$$

if and only if the algebraic content-reachable joint predictive quotient strictly refines the behavioral quotient.

Under the additional point-preparation, predictive-law, and attainability assumptions stated below, this is equivalent to the existence of an actual reproducibly attainable early predictive fiber.

Within an oracle-known analytic transfer family with fixed McMillan degree and isolated simple poles, an exact branch-opening index $\Omega_c^{\text{an}} > 0$ identifies a similarity-invariant early-neural/late-behavioral transfer branch. Under these simple-pole assumptions, $\Omega_c^{\text{an}} > 0$ already implies $q_c(\rho_0) > 0$; the rank is retained because it measures the total early relative-observability reserve rather than because it is logically independent of the qualifying branch.

Finite-stage hypothesis

Let $\mathfrak{A}_{1-\delta}$ contain every fixed-degree branch history consistent with:

- simultaneous $1 - \delta$ confidence sets for the sampled transfer functions;
- all admissible pole matchings;

- independently justified pole and residue Lipschitz bounds;
- spectral and projector-stability requirements;
- fixed input and output coordinates;
- a prespecified output metric; and
- prespecified behavioral-silence, behavioral-opening, and neural-presence margins.

Define

$$\Omega_c^- = \min_{\theta \in \mathfrak{A}_{1-\delta}} \Omega_c(\theta), \quad \Omega_c^+ = \max_{\theta \in \mathfrak{A}_{1-\delta}} \Omega_c(\theta).$$

Then:

- $\Omega_c^- > 0$ is an uncertainty-qualified certificate within the declared transport class;
- $\Omega_c^- = 0 < \Omega_c^+$ means the sampled data leave transport and nontransport histories observationally equivalent;
- $\mathfrak{A}_{1-\delta} = \emptyset$ rejects the declared fixed-degree regular transport class; and
- acceptance of a transport class does not by itself reject late-creation, switching, or measurement-drift models outside that class.

Complementary non-identifiability hypothesis

A finite passive record under an endogenous fixed policy does not identify an all-policy covert reserve when causal assignment, controlled order, and adequate horizon are unrestricted. Randomized active data are not covered by this causal no-go claim, although short horizons and misspecified measurement models can still prevent identification.

Related work and inferential gap

ERP studies have sought candidate signatures of auditory awareness, including in cross-modal distractor-induced deafness [5]. Experimental covariation between objective and subjective variables has long been identified as an obstacle to interpreting such signatures [6]. The need for neurobiological evidence does not make a named component causally or phenomenally specific [7].

Advanced EEG and neuroimaging have revealed preserved networks or covert command-following in some behaviorally unresponsive patients. A review estimates cognitive motor dissociation in up to 15–20% of patients with disorders of consciousness and reports that intensive-care detection is associated with functional recovery at one year after injury [1]. A prospective study of 16 patients with acute severe traumatic brain injury also reported task-based EEG or fMRI responses before corresponding bedside expression, while distinguishing active command-following from passive higher-order cortical responses [19]. These findings motivate longitudinal analysis but do not establish transfer of a particular early mode.

Block-Hankel realization, minimal-order inference, and partial realization form the mathematical background [41] [42] [44]. The available records are too narrow to establish novelty for the component mathematics. A recent observable-quotient proposal also treats action-conditioned observational equivalence as a basis for state abstraction, but it concerns world-model representations rather than longitudinal neural-versus-behavioral relative observability [43]. The proposed contribution is therefore presented as a synthesis and empirical specification, not as a claim to have originated Hankel realization, Kalman decomposition, spectral projectors, or modal tracking.

Formal Model

Stage coordinate and analysis unit

The principal estimand is individual-level. Each person has stages

$$\rho_0 < \rho_1 < \cdots < \rho_K$$

defined independently of the behavioral outcome. A population model may describe variation in individual parameters, but averaging kernels across patients does not produce an individual RLQO certificate.

Within each stage, parameters are treated as approximately constant over a prespecified estimation window. Recovery time, arousal, medication epoch, and clinical score are not interchangeable. If a latent disease-time coordinate is used, its alignment model must be estimated independently of the behavioral opening.

Stage-local controlled system

For discrete time $t \in \mathbb{N}_0$, let

$$x_{t+1} = A_\rho x_t + b_\rho^c \xi_t + B_\rho^n u_t^n + w_t,$$

$$Y_t^N = C_\rho^N x_t + D_\rho^N \xi_t + E_\rho^N u_t^n + v_t,$$

$$Y_t^B = C_\rho^B x_t + D_\rho^B \xi_t + E_\rho^B u_t^n + e_t.$$

Here:

- $x_t \in \mathbb{R}^{n_\rho}$ is a latent state;
- $\xi_t \in \mathbb{R}$ is the prespecified scalar content contrast;
- $u_t^n \in \mathbb{R}^{m_n}$ contains controlled or identified nuisance inputs;
- $Y_t^N \in \mathbb{R}^{p_N}$ contains measured neural outputs;
- $Y_t^B \in \mathbb{R}^{p_B}$ contains selected behavioral, communicative, or physiological outputs;
- w_t, v_t, e_t are process and measurement disturbances.

If an original multichannel content input is $u_t^c \in \mathbb{R}^{m_c}$, the scalar contrast is specified before analysis by a fixed vector a_c :

$$u_t^c = a_c \xi_t, \quad b_\rho^c = B_\rho^c a_c.$$

This restriction prevents a persistent pole that rotates from one input coordinate to another from being called persistence of the same contrast. An exploratory multichannel extension would require a prespecified geometry for residue row spaces and principal-angle margins; that extension is not proved here.

Define

$$Y_t^J = \begin{bmatrix} Y_t^N \\ Y_t^B \end{bmatrix}, \quad C_\rho^J = \begin{bmatrix} C_\rho^N \\ C_\rho^B \end{bmatrix}.$$

The direct terms D_ρ^N and D_ρ^B describe lag-zero effects. They are estimated and tested separately but excluded from the persistent-mode rank below. They must not be converted into delay states and then interpreted as neural reserve without scientific justification.

Assumptions and their roles

- **A1 — Prespecified stages:** stage times and stopping rules are not selected using the behavioral opening being tested.
- **A2 — Stage locality:** system parameters are approximately constant within each estimation window.
- **A3 — Causal contrast:** ξ is randomized or identified by a defensible causal design.
- **A4 — Stable contrast semantics:** the physical intervention and its coordinate remain aligned across stages.
- **A5 — Nuisance separation:** content, arousal, medication, timing, task load, and other nuisance effects are independently varied or identified by a valid model.
- **A6 — Reset or carryover control:** initial-state distributions are comparable, conditioned, or explicitly modeled; slow carryover is not attributed to the impulse response.
- **A7 — Finite dynamic order:** the contrast-reachable dynamic subsystem is finite-dimensional within each stage.
- **A8 — Sufficient excitation and horizon:** the experimental design spans the required input lags, and Hankel horizons reach the relevant observability and reachability indices.
- **A9 — Joint content minimality:** the contrast-reachable subsystem is observable from Y^J .
- **A10 — Predictive-law sufficiency:** for the point-state interpretation, disturbances form an identifiable state-independent translation family, so equality of conditional mean sequences is equivalent to equality of full future output laws.
- **A11 — Reproducible point preparation:** the relevant latent preparations can be generated repeatedly under admissible protocols.
- **A12 — Attainable differences:** for a fiber claim, admissible preparations realize the relevant directions in $\mathcal{U}_B/\mathcal{U}_J$.
- **A13 — Measurement alignment:** neural sensors, references, source models, preprocessing, behavioral definitions, and observation filters are calibrated or explicitly modeled.
- **A14 — Fixed measurement metric:** outputs are noise-whitened or evaluated in a prespecified metric before singular-value and residue thresholds are applied.
- **A15 — Fixed transport order:** longitudinal transport analysis assumes a constant content-reachable joint McMillan degree over the modeled interval.
- **A16 — Regular transport:** relevant poles are simple, form isolated real spectral units, and satisfy externally justified drift, residue, and projector-stability bounds.
- **A17 — Policy comparability:** behavioral predictive equivalence uses common future probes that do not adapt to neural measurements unavailable to the selected behavioral system.

A3–A6 and A13 are causal or measurement assumptions. A7–A9 concern exact realization. A10–A12 are needed only to map the algebraic quotient to actual predictive fibers. A15–A16 define the longitudinal transport model and are not consequences of finite observations.

Point-state predictive equivalence

Let $\mathcal{A}_c(\rho)$ be the set of reproducibly attainable point states at stage ρ under admissible variations of ξ , with nuisance conditions fixed or independently randomized.

For $x, x' \in \mathcal{A}_c(\rho)$, define

$$x \sim_{\rho}^J x'$$

when all common admissible future probes induce the same full future joint-output law from x and x' . Define

$$x \sim_{\rho}^B x'$$

analogously using only Y^B .

Joint equivalence implies behavioral equivalence, so there is a canonical map between the corresponding sets of attainable predictive classes:

$$\pi_{\rho} : \mathcal{A}_c(\rho) / \sim_{\rho}^J \longrightarrow \mathcal{A}_c(\rho) / \sim_{\rho}^B.$$

An attainable behavioral fiber is nontrivial when it contains two distinct joint predictive classes. These quotient sets are not assumed to be vector spaces. The linear quotient interpretation is established only after imposing A10–A12.

Dynamic interventional kernels

For $Y \in \{N, B, J\}$ and lag $k \geq 1$, define

$$G_{c,\rho}^Y(k) = \frac{\partial}{\partial a} \mathbb{E} [Y_{t+k} \mid \text{do}(\xi_t = a), \text{do}(\xi_{s \neq t} = 0), \text{do}(u^n = \bar{u}^n)] \Big|_{a=0}.$$

For a categorical contrast, the derivative is replaced by a prespecified finite randomized contrast. In the linear model,

$$G_{c,\rho}^Y(k) = C_{\rho}^Y A_{\rho}^{k-1} b_{\rho}^c.$$

The lag-zero effect is D_{ρ}^Y and is excluded from this sequence.

For positive row and column horizons r, s , define

$$\mathcal{H}_{c,\rho}^Y(r, s) = \begin{bmatrix} G_{c,\rho}^Y(1) & G_{c,\rho}^Y(2) & \cdots & G_{c,\rho}^Y(s) \\ G_{c,\rho}^Y(2) & G_{c,\rho}^Y(3) & \cdots & G_{c,\rho}^Y(s+1) \\ \vdots & \vdots & \ddots & \vdots \\ G_{c,\rho}^Y(r) & G_{c,\rho}^Y(r+1) & \cdots & G_{c,\rho}^Y(r+s-1) \end{bmatrix}.$$

It factors as

$$\mathcal{H}_{c,\rho}^Y(r, s) = \mathcal{O}_{\rho,r}^Y \mathcal{K}_{\rho,s}^c,$$

with

$$\mathcal{O}_{\rho,r}^Y = \begin{bmatrix} C_\rho^Y \\ C_\rho^Y A_\rho \\ \vdots \\ C_\rho^Y A_\rho^{r-1} \end{bmatrix}, \quad \mathcal{K}_{\rho,s}^c = \begin{bmatrix} b_\rho^c & A_\rho b_\rho^c & \cdots & A_\rho^{s-1} b_\rho^c \end{bmatrix}.$$

Define

$$q_c(\rho; r, s) = \text{rank } \mathcal{H}_{c,\rho}^J(r, s) - \text{rank } \mathcal{H}_{c,\rho}^B(r, s).$$

When both ranks have saturated,

$$q_c(\rho) = \text{rank } \mathcal{H}_c^J(\rho) - \text{rank } \mathcal{H}_c^B(\rho).$$

At population level, $\mathcal{H}_c^B$ is a row submatrix of $\mathcal{H}_c^J$, up to row permutation. Therefore

$$q_c(\rho) \geq 0.$$

Empirical joint and behavioral kernels must be estimated in one nested model. Independently regularized estimates need not preserve this relation and can manufacture a rank difference.

Transfer functions and real spectral units

The strictly proper dynamic transfer function is

$$T_{c,\rho}^Y(z) = C_\rho^Y (zI - A_\rho)^{-1} b_\rho^c.$$

For an isolated simple eigenvalue $\lambda_j(\rho)$, define the Riesz projector over the complexification:

$$P_j(\rho) = \frac{1}{2\pi i} \oint_{\Gamma_j(\rho)} (zI - A_\rho)^{-1} dz.$$

Its output residue is

$$R_j^Y(\rho) = C_\rho^Y P_j(\rho) b_\rho^c.$$

For a real system, a nonreal pole and its conjugate represent one real two-dimensional oscillatory unit. RLQO therefore counts:

- one unit for a simple real pole; and
- one unit for a simple nonreal conjugate pair, represented by the pole in the open upper half-plane.

Conjugate poles are not counted twice. A unit count is not itself a state dimension; the corresponding real dimensions are one and two, respectively.

Define neural and behavioral residue magnitudes in the prespecified output metric:

$$n_j(\rho) = \|R_j^N(\rho)\|, \quad b_j(\rho) = \|R_j^B(\rho)\|.$$

For a conjugate pair, either member has the same magnitude. Exact behavioral silence means the complex residue is zero, which is equivalent to zero contribution from the associated real invariant plane.

Oracle-level analytic opening index

Suppose an analytic fixed-order transfer family is known over the entire interval and all relevant real spectral units remain isolated. Define

$$\Omega_c^{\text{an}}(\rho_0, \rho_1) = \sum_j \mathbf{1} \left[\inf_{\rho \in [\rho_0, \rho_1]} n_j(\rho) > 0, b_j(\rho_0) = 0, b_j(\rho_1) > 0 \right].$$

This is a structural index of a supplied continuous transfer family. It is not identifiable from finitely many stage estimates without additional regularity bounds.

Finite-stage uncertainty-qualified transport

Let $\mathcal{C}_k$ be a simultaneous confidence region for the stage- k joint transfer function or an equivalent bounded-order parameterization. Let $\mathfrak{A}_{1-\delta}$ be the set of all longitudinal transfer families θ satisfying:

1. $T_c^\theta(\rho_k) \in \mathcal{C}_k$ for every k ;
2. constant joint McMillan degree n_c ;
3. simple real spectral units with prespecified minimum separation and projector conditioning;
4. all pole and residue trajectories obey externally justified Lipschitz bounds;
5. all permutations of poles between stages consistent with those bounds are retained;
6. the contrast and output metrics remain fixed; and
7. the observation model belongs to the prespecified calibrated family.

Let $\nu > 0$ be a neural-presence margin, $\alpha \geq 0$ an early behavioral-equivalence margin, and $\beta > \alpha$ a later opening margin. For $\theta \in \mathfrak{A}_{1-\delta}$, define

$$\Omega_c(\theta) = \sum_j \mathbf{1} \left[\inf_{\rho \in [\rho_0, \rho_K]} n_j^\theta(\rho) \geq \nu, b_j^\theta(\rho_0) \leq \alpha, b_j^\theta(\rho_K) \geq \beta \right].$$

Then define the partial-identification interval

$$\boxed{\Omega_c^- = \min_{\theta \in \mathfrak{A}_{1-\delta}} \Omega_c(\theta), \quad \Omega_c^+ = \max_{\theta \in \mathfrak{A}_{1-\delta}} \Omega_c(\theta).}$$

If the set is nonempty:

- $\Omega_c^- > 0$ means every admissible transport history contains at least one qualifying unit;
- $\Omega_c^- = 0 = \Omega_c^+$ excludes such a unit within the class;
- $\Omega_c^- = 0 < \Omega_c^+$ leaves qualifying and nonqualifying histories observationally equivalent; and
- $\Omega_c^- = \Omega_c^+$ identifies the number of qualifying units within the declared class.

This definition propagates uncertainty over branch matchings rather than choosing one best match and treating it as known.

Neuroscientific Integration

Cue, operation, and inferred content

A randomized command cue is not a direct intervention on comprehension, imagery, intention, or conscious content. At least six variables should remain distinct:

1. physical cue identity;
2. instructed cognitive operation;
3. inferred internal content;
4. task compliance;
5. neural classifier output; and
6. overt or assisted communication.

A stronger design varies cue form and intended operation orthogonally. Examples include multiple cues mapping to the same intended operation, multiple operations presented in the same sensory format, negative-control linguistic contrasts, and manipulation checks for sensory processing. Without such controls, $q_c > 0$ may reflect an unconscious cue trace.

ERP measurements are projections of a multichannel response family onto selected sensors, references, filters, and latency windows. A component contrast can be useful without identifying system order or a persistent branch. In a cross-modal auditory paradigm, a possible mapping is:

Formal object	Possible realization
ξ	prespecified randomized target or instruction contrast
u^n	distractor load, cue timing, modality, vigilance, medication, or task demand
Y^N	multichannel evoked EEG or source-resolved activity
Y^B	detection, discrimination, confidence, command-following, or communication
D^Y	instantaneous or unresolved stimulus-locked response
$G_c^Y(k)$	dynamic contrast-specific response after lag zero

ERP work on cross-modal distractor-induced deafness supplies a relevant paradigm, not a validated RLQO marker [5]. Covariation of objective and subjective variables instead motivates independent manipulation of the contrast, nuisance factors, and output demands [6].

Disorders of consciousness

Task-based EEG and fMRI can reveal active command-following or passive higher-order cortical responses without corresponding bedside behavior [19]. These outcomes must remain distinct. A passive association-cortex response to language or music is not equivalent to following a command.

A clinical implementation might use:

Formal object	Candidate clinical realization
ξ	one preregistered motor-imagery, spatial-navigation, linguistic, or sensory contrast
u^n	medication, sleep–wake state, autonomic state, cue timing, background stimulation, or task load
Y^N	EEG, source-resolved electrophysiology, fMRI with an explicit hemodynamic model, or perturbation-evoked activity
Y^B	a separately validated command-following, communication, gaze, or choice channel
ρ_k	preregistered within-person calendar times or clinical epochs
$\mathcal{C}_k$	uncertainty region from repeated trials within stage k

Multiple tasks and modalities may reveal responses missed by one protocol because injuries and available cognitive resources are heterogeneous. The supporting study was small—14 patients initially and complete cross-task data in six—so this should be treated as feasibility evidence rather than established population-level sensitivity improvement [4].

EEG command-following signals can vary substantially across time and space, and accounting for background-state fluctuations can change classification [21]. A stage-local model must therefore include trial-level arousal covariates, prespecified exclusions, and uncertainty due to state fluctuations. Selecting the first session with a positive behavior and then asking whether a residue “opened” at that session would be outcome-conditioned and partly tautological.

Command-following and communication are different output systems. Their dissociation has been examined in task design, including in healthy-subject fMRI work [22]. A mode may be open relative to gaze but closed relative to speech, or open relative to a classifier-assisted channel but closed relative to unaided movement. Accordingly, RLQO is output-relative rather than report-independent.

Graph-theoretic EEG studies have reported associations between network organization and behavioral awareness, including relatively preserved alpha networks in some behaviorally unresponsive patients with covert-awareness evidence [2]. A graph score may be included as a measured neural output or covariate, but it does not by itself establish contrast reachability, early behavioral silence, or branch continuity.

Modality-specific observation models

The measured neural output need not share the poles of neuronal dynamics.

For fMRI, a simplified observation model is

$$Y_{\rho}^{\text{fMRI}} = F_{\rho}^{\text{HRF}}(q) Y_{\rho}^{\text{neuronal}} + \epsilon_{\rho},$$

where $F_{\rho}^{\text{HRF}}(q)$ is a stage- and patient-dependent hemodynamic filter. If this filter changes, apparent poles may move or appear even when neuronal dynamics do not.

For EEG, reference, volume conduction, source mixing, muscle activity, impedance, and preprocessing alter the observation operator. Exact rank is invariant under an invertible fixed output transformation, but thresholded rank and residue norms are not invariant to arbitrary rescaling, rank-deficient projection,

rereferencing, or changing noise covariance. Output whitening and calibration must therefore precede threshold comparisons.

Sensor and hemodynamic states may be included in an augmented model only if they are labeled as measurement dynamics. Their poles must not be interpreted as neural reserve merely because they appear in a joint realization.

Perturbational evidence

An external perturbation may be represented by

$$x_{t+1} = A_\rho x_t + b_\rho^c \xi_t + B_\rho^p u_t^p + B_\rho^n u_t^n + w_t.$$

Perturbation can constrain propagation and responsiveness, but it identifies A_ρ only when stimulation provides adequate controllability and measurements provide adequate observability. Reliability is necessary but not sufficient.

Joint use of FDG-PET and perturbational complexity produced congruent classifications in 22 of 24 participants and identified preserved metabolism and high complexity in four behaviorally unresponsive participants [28]. This small study supports multimodal complementarity, not content-specific transport.

TMS–EEG methodology remains sensitive to artifact management and analysis choices [29]. A pre-peer-review reliability review likewise emphasizes explicit reliability assessment before broad clinical translation of TMS-evoked potentials [30]. Auditory and somatosensory contamination, stimulation-target drift, and changing muscle artifacts are matched alternatives to neural branch transport.

Clinical evidential hierarchy

The framework distinguishes:

1. **Recovery correlation:** an early signal predicts later outcome.
2. **Contrast-specific response:** the randomized contrast changes a neural output.
3. **Positive relative rank:** joint dynamic order exceeds selected behavioral dynamic order.
4. **Attainable predictive fiber:** reproducible preparations differ jointly but not behaviorally.
5. **Uncertainty-qualified transport:** every admissible fixed-order branch history contains an early-neural/late-behavioral unit.
6. **Awareness or phenomenality:** an additional claim requiring independent evidence and a bridge hypothesis.

Repeated standardized examination, EEG, and functional neuroimaging are recommended as complementary elements of assessment in disorders of consciousness [3]. RLQO does not replace any of them.

Philosophical Reinterpretation

Report-hidden status as an output-relative relation

RLQO replaces the phrase “report-independent content” with **report-hidden relative to a selected output family**. Adding eye tracking, autonomic choice, assisted communication, or a brain–computer interface can change C^B , reduce q_c , and eliminate an apparent fiber.

The evidential structure is diachronic:

early contrast-sensitive joint distinction $\longrightarrow$ admissible branch histories $\longrightarrow$ later selected

Later behavior does not cause an earlier state. It only constrains retrospective interpretation. Nor does a transfer branch establish numerical identity of neurons, persistence of a biological mechanism, or identity of a phenomenal subject. It establishes compatibility with a common input–output branch inside a specified model class.

Distinct explanatory targets

The following claims are not interchangeable:

1. stimulation affects the nervous system;
2. an imposed contrast produces different neural dynamics;
3. the contrast affects a selected voluntary or physiological output;
4. the patient comprehends the contrast;
5. the patient can intentionally deploy the represented distinction;
6. the distinction is globally or flexibly accessible;
7. the distinction is phenomenally experienced.

The mathematics addresses relations among the first three. Evidence about comprehension, volition, and flexible use must be added by experimental design. Phenomenality is not represented by any state-space variable.

Let $P_\phi(\rho_0)$ mean that the system was phenomenally conscious at the early stage. RLQO does not prove

$$q_c(\rho_0) > 0 \wedge \Omega_c^- > 0 \implies P_\phi(\rho_0).$$

Any such implication requires a bridge hypothesis H_ϕ . A scientifically useful bridge would need independent scope conditions—for example, prior validation of comprehension, voluntary control, flexible generalization, and correspondence with later structured first-person reports. The non-conscious cache below is a live specificity countermodel to any bridge based only on rank and transport.

First-person evidence

Where communication is possible, Y^B may include structured reports of content, confidence, temporal order, qualitative similarity, perceptual presence, ownership, or affective significance. Linking a transported branch to such reports is more informative than linking it only to movement, but it remains a third-person model of report behavior. It does not derive qualitative character from transfer-function structure.

Affective and interoceptive boundary

Endogenous pain, emotion, bodily regulation, and interoception are not naturally represented as randomized exogenous scalar inputs. Autonomic variables may simultaneously be nuisance inputs, outputs, and constitutive state variables. The present model may be applied to controlled perturbations of those systems, but it does not identify endogenous affective experience. Theoretical proposals concerning emotional consciousness likewise do not remove this causal boundary^[13].

Evidential and ethical asymmetry

Failure to identify RLQO does not show absence of awareness. The relevant distinction may be unexcited, outside sensor coverage, nonlinear, nonstationary, inaccessible through the selected output, or present without later behavioral recovery.

Acute brain-injury research emphasizes that recovery can precede self-expression and that inaccurate prognosis can affect access to life-sustaining treatment and rehabilitation ^[19]. It is therefore defensible, as a precautionary norm rather than a mathematical theorem, to treat positive evidence as a reason for increased protective attention while refusing to treat negative RLQO evidence as exclusionary.

Theorem, Proposition, Proof, Derivation, Counterexample, or No-Go Result

Reachable and unobservable subspaces

Fix a stage and suppress ρ . Define the scalar-contrast-reachable subspace

$$\mathcal{R}_c = \text{span} \{b^c, Ab^c, A^2b^c, \dots\}.$$

For $Y \in \{B, J\}$, define

$$\mathcal{U}_Y = \{x \in \mathcal{R}_c : C^Y A^k x = 0 \text{ for every } k \geq 0\}.$$

Because the joint output contains the behavioral output,

$$\mathcal{U}_J \subseteq \mathcal{U}_B.$$

Under joint observability on $\mathcal{R}_c$,

$$\mathcal{U}_J = \{0\}.$$

Theorem 1 — Relative Hankel-rank identity

Statement. Choose s such that

$$\text{im } \mathcal{K}_s^c = \mathcal{R}_c$$

and r such that, for $Y = B, J$,

$$\ker(\mathcal{O}_r^Y|_{\mathcal{R}_c}) = \mathcal{U}_Y.$$

Then

$$\boxed{\text{rank } \mathcal{H}_c^J(r, s) - \text{rank } \mathcal{H}_c^B(r, s) = \dim \mathcal{U}_B - \dim \mathcal{U}_J = \dim(\mathcal{U}_B/\mathcal{U}_J).}$$

If the contrast-reachable realization is jointly observable, then

$$q_c = \dim \mathcal{U}_B.$$

Proof. The factorization

$$\mathcal{H}_c^Y(r, s) = \mathcal{O}_r^Y \mathcal{K}_s^c$$

and surjectivity of $\mathcal{K}_s^c$ onto $\mathcal{R}_c$ imply

$$\text{rank } \mathcal{H}_c^Y(r, s) = \text{rank}(\mathcal{O}_r^Y|_{\mathcal{R}_c}).$$

By rank–nullity,

$$\text{rank}(\mathcal{O}_r^Y|_{\mathcal{R}_c}) = \dim \mathcal{R}_c - \dim \ker(\mathcal{O}_r^Y|_{\mathcal{R}_c}).$$

The horizon assumption gives

$$\text{rank } \mathcal{H}_c^Y(r, s) = \dim \mathcal{R}_c - \dim \mathcal{U}_Y.$$

Subtracting the behavioral rank from the joint rank yields

$$\text{rank } \mathcal{H}_c^J - \text{rank } \mathcal{H}_c^B = \dim \mathcal{U}_B - \dim \mathcal{U}_J.$$

Since $\mathcal{U}_J \subseteq \mathcal{U}_B$,

$$\dim \mathcal{U}_B - \dim \mathcal{U}_J = \dim(\mathcal{U}_B/\mathcal{U}_J).$$

Under joint observability, $\mathcal{U}_J = \{0\}$, giving $q_c = \dim \mathcal{U}_B$. $\square$

This theorem concerns the linear contrast-reachable span. It does not by itself show that every quotient direction is reproducibly attainable under bounded, categorical, or policy-constrained inputs.

Corollary 1 — Algebraic and attainable predictive quotients

Assume A10 and common future probes. On $\mathcal{R}_c$, joint and behavioral point-state predictive quotients are represented by

$$\mathcal{R}_c/\mathcal{U}_J \quad \text{and} \quad \mathcal{R}_c/\mathcal{U}_B.$$

The canonical linear map

$$\bar{\pi} : \mathcal{R}_c/\mathcal{U}_J \longrightarrow \mathcal{R}_c/\mathcal{U}_B, \quad [x]_{\mathcal{U}_J} \mapsto [x]_{\mathcal{U}_B},$$

has

$$\ker \bar{\pi} \cong \mathcal{U}_B/\mathcal{U}_J.$$

Therefore,

$$q_c = \dim \ker \bar{\pi}.$$

If, in addition, $\mathcal{A}_c$ is an affine set with

$$\mathcal{A}_c - \mathcal{A}_c = \mathcal{R}_c,$$

or more generally realizes every nonzero class in $\mathcal{U}_B/\mathcal{U}_J$, then

$$q_c > 0$$

if and only if the attainable map π has a nontrivial fiber.

Proof. For point preparations x, x' , equality of future behavioral laws is, by A10, equivalent to equality of conditional mean sequences:

$$C^B A^k (x - x') = 0 \quad \text{for every } k \geq 0.$$

Thus $x - x' \in \mathcal{U}_B$. The joint condition similarly gives $x - x' \in \mathcal{U}_J$. The quotient representations follow.

The kernel of $\bar{\pi}$ consists of classes represented by $\mathcal{U}_B$, modulo $\mathcal{U}_J$, so

$$\ker \bar{\pi} \cong \mathcal{U}_B/\mathcal{U}_J.$$

Theorem 1 gives its dimension.

If an attainable fiber is nontrivial, two attainable points differ by an element of $\mathcal{U}_B \setminus \mathcal{U}_J$, so $q_c > 0$. Conversely, if $q_c > 0$ and the attainable-set condition realizes a nonzero quotient class, two reproducible preparations occupy different joint classes but the same behavioral class. $\square$

Without the attainable-set and predictive-law assumptions, the last equivalence is not valid. Persistent excitation of a linear span is not a substitute for reproducible state preparation.

Corollary 2 — Similarity invariance

Let S be invertible and define

$$A' = SAS^{-1}, \quad b^{c'} = Sb^c, \quad C^{Y'} = C^Y S^{-1}.$$

Then

$$C^{Y'} (A')^{k-1} b^{c'} = C^Y A^{k-1} b^c$$

for every $k \geq 1$. Thus all dynamic Markov parameters, Hankel matrices, exact ranks, and values of q_c are unchanged.

Moreover,

$$\mathcal{R}'_c = S\mathcal{R}_c, \quad \mathcal{U}'_Y = S\mathcal{U}_Y,$$

so the quotient dimension is invariant. This does not make approximate ranks invariant to arbitrary output rescaling or changing measurement operators.

Constructive derivation — Exact minimal realization

Let

$$n_c = \text{rank } \mathcal{H}_c^J$$

and choose an exact rank factorization

$$\mathcal{H}_c^J = LR,$$

where L has n_c independent columns and R has n_c independent rows. Define the shifted Hankel matrix

$$\mathcal{H}_c^{J,+} = \begin{bmatrix} G_c^J(2) & \cdots & G_c^J(s+1) \\ \vdots & \ddots & \vdots \\ G_c^J(r+1) & \cdots & G_c^J(r+s) \end{bmatrix}.$$

Set

$$\hat{A} = L^\dagger \mathcal{H}_c^{J,+} R^\dagger.$$

Let $\hat{b}^c$ be the first column of R , and let $\hat{C}^J$ be the first $p_N + p_B$ rows of L .

For an exact minimal realization,

$$\mathcal{H}_c^J = \mathcal{O}_r^J \mathcal{K}_s^c.$$

Any two full-rank factorizations differ by an invertible S :

$$L = \mathcal{O}_r^J S, \quad R = S^{-1} \mathcal{K}_s^c.$$

Since

$$\mathcal{H}_c^{J,+} = \mathcal{O}_r^J A \mathcal{K}_s^c,$$

it follows that

$$\hat{A} = S^{-1} A S, \quad \hat{b}^c = S^{-1} b^c, \quad \hat{C}^J = C^J S.$$

Thus exact finite-order kernels determine a minimal realization up to similarity. This is an algebraic derivation, not a finite-sample consistency or stability theorem.

Lemma — A silent simple spectral unit supplies a relative-observability direction

Suppose a simple real pole has projector P_j and satisfies

$$C^N P_j b^c \neq 0, \quad C^B P_j b^c = 0.$$

Then its reachable modal line is neurally observable and behaviorally unobservable.

For a nonreal conjugate pair, if the residue at the upper-half-plane pole is neurally nonzero and behaviorally zero, the associated real two-dimensional invariant plane contains a contrast-reachable neurally observable component and is behaviorally unobservable.

Proof. For a simple pole, P_j has complex rank one. Neural nonzero residue implies $P_j b^c \neq 0$. Writing $P_j = v_j w_j^*$, reachability gives $w_j^* b^c \neq 0$. Behavioral zero residue then forces

$$C^B v_j = 0.$$

Because $A v_j = \lambda_j v_j$,

$$C^B A^k v_j = \lambda_j^k C^B v_j = 0$$

for every k , whereas the neural residue is nonzero.

For a nonreal pole, the same argument applies over the complexification. Since $C^B v_j = 0$, both the real and imaginary parts are behaviorally unobservable. They span the associated real invariant plane unless the mode degenerates, which is excluded by simplicity. $\square$

Theorem 2 — Conditional analytic branch invariance

Statement. Let $\rho \in [\rho_0, \rho_1]$. Assume:

1. the contrast-reachable joint McMillan degree is constant;
2. an aligned minimal realization of that fixed dimension has matrices analytic in ρ ;
3. the relevant poles remain simple and isolated;
4. input and output coordinates retain their meanings; and
5. nonreal conjugate pairs are grouped as real spectral units.

Then:

1. each relevant pole defines an analytic branch while isolation persists;
2. each pole residue is determined by the transfer matrix and is invariant under latent-state similarity;
3. the exact zero/nonzero classification in Ω_c^{an} is invariant across minimal realizations of the same analytic transfer family; and
4. Ω_c^{an} counts real spectral units without double-counting conjugate poles.

Proof. In a minimal realization, transfer poles are precisely the reachable and observable eigenvalues; a pole-zero cancellation would contradict minimality. Around a simple pole,

$$T_{c,\rho}^Y(z) = \frac{R_j^Y(\rho)}{z - \lambda_j(\rho)} + T_{j,\rho}^{Y,\text{reg}}(z).$$

Uniqueness of the Laurent expansion makes λ_j and R_j^Y transfer-function invariants.

Analytic dependence and spectral isolation permit a continuously deformed contour around the pole. The Riesz integral therefore defines an analytic projector branch. Under a stage-wise similarity,

$$A'_\rho = S_\rho A_\rho S_\rho^{-1}, \quad b_{\rho'}^{c'} = S_\rho b_\rho^c, \quad C_{\rho'}^{Y'} = C_\rho^Y S_\rho^{-1},$$

the projector transforms as

$$P'_j = S_\rho P_j S_\rho^{-1}.$$

Hence

$$C_\rho^{Y'} P'_j b_\rho^{c'} = C_\rho^Y P_j b_\rho^c.$$

Residues and their exact zero/nonzero status are invariant. For real systems, conjugate residues carry the same zero/nonzero classification, so grouping the pair produces one real spectral unit. $\square$

This theorem assumes the continuous branch whose existence is under discussion. It does not show that finitely sampled stages determine a unique analytic interpolation, exclude an unobserved residue zero, or reject mode creation.

Proposition 1 — Corrected operational characterization

Let $\text{RLQO}_c^{\text{an}}(\rho_0, \rho_1)$ mean that:

1. there is a nontrivial reproducibly attainable early predictive fiber; and
2. one simple real spectral unit in that fiber satisfies the exact neural-presence and behavioral-opening conditions of Ω_c^{an} .

Under Theorem 1, Corollary 1 including its attainability and predictive-law assumptions, and Theorem 2,

$$\boxed{\text{RLQO}_c^{\text{an}}(\rho_0, \rho_1) \iff q_c(\rho_0) > 0 \wedge \Omega_c^{\text{an}}(\rho_0, \rho_1) > 0.}$$

Proof. If $\text{RLQO}_c^{\text{an}}$ holds, the attainable fiber supplies a direction in $\mathcal{U}_B \setminus \mathcal{U}_J$, so $q_c > 0$. Its qualifying unit contributes to Ω_c^{an} .

Conversely, $\Omega_c^{\text{an}} > 0$ supplies, by the lemma, an early contrast-reachable direction that is neurally observable and behaviorally unobservable. The attainable-set assumption realizes the corresponding predictive distinction. Theorem 2 supplies branch identity within the analytic model, and the late nonzero behavioral residue supplies opening. $\square$

The conjunction is not two independent exact tests: under the simple-pole assumptions, positive Ω_c^{an} already implies positive q_c . The rank remains useful as a summary of all early report-hidden linear dimensions and as a separate noisy-data diagnostic.

Proposition 2 — Deterministic singular-value robustness

For a matrix M , define

$$r_\varepsilon(M) = \#\{i : \sigma_i(M) > \varepsilon\}.$$

Suppose

$$\|\widehat{\mathcal{H}}_c^Y - \mathcal{H}_c^Y\|_2 \leq \delta_Y, \quad Y \in \{J, B\}.$$

Let γ_Y be the smallest nonzero singular value of $\mathcal{H}_c^Y$. If

$$\delta_Y < \varepsilon_Y < \gamma_Y - \delta_Y,$$

then

$$r_{\varepsilon_Y}(\widehat{\mathcal{H}}_c^Y) = \text{rank } \mathcal{H}_c^Y.$$

If $\mathcal{H}_c^Y = 0$, it is sufficient that $\varepsilon_Y > \delta_Y$.

Proof. Singular-value perturbation yields

$$\left| \sigma_i(\widehat{\mathcal{H}}_c^Y) - \sigma_i(\mathcal{H}_c^Y) \right| \leq \delta_Y.$$

Population-zero singular values remain below ε_Y , while population-nonzero singular values remain above it. $\square$

A separating threshold can exist only if

$$\gamma_Y > 2\delta_Y.$$

The proposition assumes valid error bounds. It supplies neither a null distribution nor a method for choosing ε_Y . Because behavioral rows are nested in the joint Hankel matrix, the preferred inferential object is a joint confidence set constrained by that nesting, not two unrelated thresholded estimates.

Proposition 3 — Finite-stage partial-identification validity

Assume that, with probability at least $1 - \delta$, the simultaneous confidence regions contain all true sampled transfer functions. Also assume that the true longitudinal transfer family satisfies the fixed-degree, metric, separation, and Lipschitz restrictions defining $\mathfrak{A}_{1-\delta}$.

Then, with probability at least $1 - \delta$,

$$\Omega_c^- \leq \Omega_c^{\text{true}} \leq \Omega_c^+.$$

Consequently,

$$\Omega_c^- > 0$$

is a level- δ , model-conditional certificate that the true admissible transport family contains a qualifying branch.

Proof. On the simultaneous coverage event, the true sampled transfer functions belong to all $\mathcal{C}_k$. By the regularity assumptions, the true longitudinal family is therefore an element of $\mathfrak{A}_{1-\delta}$. The minimum and maximum over that set bound its value of Ω_c . $\square$

The result is conditional on the true family belonging to the declared transport class. It does not assign coverage to an incorrect order bound, an unmodeled switch, or a changing observation filter.

Proposition 4 — Sufficient discrete matching and interstage-presence bounds

Suppose adjacent stages are separated by

$$\Delta_k = \rho_{k+1} - \rho_k.$$

Let L_λ be an independently justified pole Lipschitz constant, so the true pole drift is at most

$$d_k = L_\lambda \Delta_k.$$

Let g_k be the minimum true pole separation at stage k , and let each estimated pole have error at most η_k . Nearest-neighbor matching is unique if

$$d_k + 2\eta_k < \frac{g_k}{2}.$$

Proof. The estimated distance to the true continuation is at most $d_k + 2\eta_k$. A competing pole is at estimated distance at least $g_k - d_k - 2\eta_k$. The former is smaller exactly when

$$d_k + 2\eta_k < g_k - d_k - 2\eta_k,$$

which is equivalent to the stated condition. $\square$

This condition is sufficient, not necessary. If it fails, all compatible matchings must remain in $\mathfrak{A}_{1-\delta}$.

For continuous neural presence, suppose simultaneous samplewise lower bounds satisfy

$$n_j(\rho_k) \geq \underline{n}_{j,k},$$

and n_j is L_N -Lipschitz. On $[\rho_k, \rho_{k+1}]$,

$$n_j(\rho) \geq \max \{ \underline{n}_{j,k} - L_N(\rho - \rho_k), \underline{n}_{j,k+1} - L_N(\rho_{k+1} - \rho) \}.$$

If the infimum of this lower envelope exceeds ν on every interval, an unobserved neural-residue zero is excluded within the class. Analyticity alone supplies no such finite-sample guarantee.

Similarly, early silence and late opening require confidence bounds of the form

$$\bar{b}_j(\rho_0) \leq \alpha, \quad \underline{b}_j(\rho_K) \geq \beta, \quad \beta > \alpha.$$

Failure to reject a point null of zero behavioral residue is not evidence of silence.

Theorem 3 — Passive fixed-policy no-go result

Statement. Let D_H be a finite-horizon distribution of observed assignments, nuisance variables, neural outputs, and behavioral outputs under one passive endogenous policy through horizon H . Consider a broad controlled-model class that permits finite shift-register states, random initial conditions storing passive trajectories, and no fixed upper bound on delay order. Suppose:

1. causal effects of the assigned contrast are not independently identified;
2. no justified upper bound on controlled order is imposed; and
3. counterfactual policies and responses beyond H are untested.

Then D_H does not identify whether a pre-existing contrast-sensitive mode is neurally present and later becomes behaviorally effective.

Proof. Construct M_0 as a passive replay system. A random initial state stores a complete trajectory sampled from D_H , together with a clock. Under the observed policy, a time-invariant shift register emits the stored assignments and outputs in sequence. Under interventions that override the observed contrast, define the emitted neural and behavioral outputs to remain those stored in the initial state. Thus M_0 reproduces D_H exactly, but its interventional contrast kernels vanish.

Construct M_1 by adjoining a delay line:

$$\begin{aligned} d_{t+1}^0 &= \xi_t, \\ d_{t+1}^i &= d_t^{i-1}, \quad i = 1, \dots, H, \\ h_{t+1} &= \lambda h_t + d_t^H, \quad 0 < |\lambda| < 1. \end{aligned}$$

Add

$$Y_t^{N,\text{add}} = \ell_N h_t, \quad Y_t^{B,\text{add}} = g(\rho) \ell_B h_t,$$

with $\ell_N, \ell_B \neq 0$,

$$g(\rho_0) = 0, \quad g(\rho_1) \neq 0.$$

The added subsystem is initialized at zero. A contrast delivered during the observed interval cannot reach h_t within horizon H , so the added outputs vanish throughout the record. Hence M_0 and M_1 induce the same D_H .

Beyond the observed horizon, M_1 contains a contrast-reachable persistent pole. It is neurally observable and behaviorally silent at ρ_0 , then behaviorally effective at ρ_1 . M_0 contains no corresponding interventional response. $\square$

The theorem applies to the broad class just stated. It does not establish impossibility in a known-order randomized LTI experiment, and it does not show that adding any intervention automatically produces identification.

Countermodel or Null System

Common-gain and weak-behavior null

Let

$$z_{t+1} = \lambda z_t + \xi_t, \quad |\lambda| < 1,$$

and let

$$Y_t^N = g_N(\rho) z_t, \quad Y_t^B = g_B(\rho) z_t,$$

where both gains are nonzero at every stage but $g_B(\rho_0)$ is small.

Then

$$G_c^N(k) = g_N(\rho)\lambda^{k-1}, \quad G_c^B(k) = g_B(\rho)\lambda^{k-1}.$$

Both Hankel matrices have rank one:

$$\text{rank } \mathcal{H}_c^J = \text{rank } \mathcal{H}_c^B = 1,$$

so

$$q_c = 0.$$

The branch is behaviorally weak, not behaviorally unobservable. A conventional significance test may fail to detect the early output, but an equivalence test with a clinically meaningful α need not classify it as silent.

Content-independent motor-recovery null

Let

$$m_{t+1} = \beta m_t + b_n u_t^n,$$

$$Y_t^N = c_N m_t, \quad Y_t^B = c_B(\rho) m_t,$$

with no contrast input:

$$b^c = 0.$$

Then

$$\mathcal{H}_c^J = \mathcal{H}_c^B = 0, \quad q_c = 0, \quad \Omega_c = 0.$$

Motor readiness may recover and covary with behavior, but it cannot satisfy a contrast-specific certificate unless contrast and nuisance channels are confounded.

Counterexample — Pole persistence without persistence of the same contrast

Consider a two-channel content input $u_t = (u_{1t}, u_{2t})^\top$:

$$x_{t+1} = \lambda x_t + \begin{bmatrix} \cos \theta(\rho) & \sin \theta(\rho) \end{bmatrix} u_t,$$

$$Y_t^N = x_t, \quad Y_t^B = g(\rho) x_t.$$

Let

$$\theta(\rho_0) = 0, \quad \theta(\rho_1) = \frac{\pi}{2},$$

and let $g(\rho_0) = 0, g(\rho_1) = 1$.

The pole λ and a nonzero MIMO residue persist, but the mode is driven by channel 1 initially and channel 2 later. An index asking only whether the pole remains sensitive to *some* content channel would count it, even though the input loading changes completely.

For the fixed contrast $c = e_1$, the later residue is zero, so the revised contrast-specific Ω_c does not count the branch. Stable labels for input coordinates are therefore insufficient; the transport claim must be contrast-specific or must impose an explicit loading-overlap geometry.

Positive non-conscious cache

Let

$$h_{t+1} = \lambda h_t + \xi_t, \quad 0 < |\lambda| < 1,$$

$$Y_t^N = h_t, \quad Y_t^B = g(\rho)h_t,$$

with

$$g(\rho_0) = 0, \quad g(\rho_1) = 1.$$

At the early stage,

$$\text{rank } \mathcal{H}_c^J = 1, \quad \text{rank } \mathcal{H}_c^B = 0,$$

so

$$q_c(\rho_0) = 1.$$

The pole is λ , with

$$R^N(\rho) = 1, \quad R^B(\rho_0) = 0, \quad R^B(\rho_1) = 1.$$

Thus the exact analytic certificate is positive.

Now stipulate that h_t is an electronic cache with a monitoring line and a recovering output gate, and that the cache is non-conscious. The equations and all RLQO quantities are unchanged. Therefore,

$$q_c > 0 \wedge \Omega_c > 0 \setminus \text{centernot} \implies \text{phenomenal consciousness.}$$

This is a logical countermodel to phenomenal sufficiency, not an empirical theory of which physical systems are conscious.

Newly created or replaced late mode

Suppose the late transfer function is

$$T_{c,\rho_1}^J(z) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \frac{1}{z - \lambda}.$$

A recovery model may create this mode only at ρ_1 , while another transports an earlier neurally visible mode. Late cross-sectional data cannot distinguish them.

A fixed-degree analytic model excludes genuine creation by assumption. Finite-stage RLQO therefore supports transport over creation only when matched late-creation, replacement, switching, and measurement-drift models are empirically rejected. If those alternatives remain compatible with held-out data, the warranted conclusion is model-class ambiguity.

Nonstationary and aggregation nulls

A mixture of stage-local systems need not itself be a stage-local finite-order system of the same dimension. For example, averaging two rank-one kernels,

$$G(k) = p\lambda_1^{k-1} + (1-p)\lambda_2^{k-1}, \quad \lambda_1 \neq \lambda_2,$$

generically produces Hankel rank two. If mixture weights differ between neural and behavioral samples because of missing trials, artifact rejection, arousal selection, or patient pooling, an apparent relative rank can arise without any individual having that rank difference.

RLQO must therefore be estimated within individuals or within an explicitly identified hierarchical model that preserves individual kernels. A group-average pole is not evidence that any patient possesses that pole.

Empirical Boundary Conditions and Falsification Criteria

Prospective design requirements

1. **Preregister stages independently of behavior.** Use calendar times or predefined windows, not the first recovered response.
2. **Declare the estimand.** Individual-level transport is primary; population summaries must preserve individual uncertainty.
3. **Randomize the contrast.** Record assignment probabilities and test compliance with the intervention schedule.
4. **Orthogonalize cue and intended operation.** Include sensory, linguistic, and motor negative controls.
5. **Control initial conditions.** Use washout, conditioning variables, or an explicit carryover model.
6. **Estimate joint and behavioral kernels together.** Preserve the row nesting of $\mathcal{H}_c^B$ in $\mathcal{H}_c^J$.
7. **Specify the output metric.** Noise-whiten or otherwise normalize measurements before thresholding.
8. **Predefine horizons and order candidates.** Report results across admissible horizons rather than selecting the most favorable one.
9. **Predefine α, β, ν .** Early silence is an equivalence claim, not a failure to reject zero.
10. **Calibrate observation dynamics.** Model hemodynamic, reference, sensor, and perturbation artifacts.
11. **Use held-out trials or stages.** Rank and modal fit must predict unfit Markov parameters or later kernels.

12. **Retain ambiguous branch matchings.** Do not force a nearest match when confidence regions overlap.

Statistical nulls

Relative-rank null

The primary nested null is

$$H_{0,q} : q_c = 0.$$

It should be fitted in a common bounded-order model for joint and behavioral kernels. Valid inferential approaches may include:

- intervention-respecting randomization tests;
- block or trial-level bootstrap procedures preserving temporal dependence;
- parametric bootstrap under the nested null;
- simultaneous confidence sets for structured low-rank kernels; or
- Bayesian posterior uncertainty under a declared order prior.

Rank is nonregular near zero singular values. No single resampling method is uniformly valid without model conditions. Results should therefore include singular spectra, horizon sensitivity, order sensitivity, and calibration simulations.

Transport null

A useful composite null is

$$H_{0,\Omega} : \text{no qualifying pre-existing branch is required by the data.}$$

It includes:

- weak but nonzero early behavior;
- a residue zero between sampled stages;
- alternative branch pairings;
- late mode creation;
- mode replacement or switching;
- a recovering relay with no continuous branch;
- changing hemodynamic or sensor filters; and
- a different late mode with a similar pole.

Testing only whether a preferred transported model fits is insufficient. Transport and alternative models should be compared under matched effective complexity, using held-out prediction and uncertainty-aware model selection. If both remain adequate, the conclusion is partial identification rather than transport.

Multiplicity and selection

Multiplicity arises from searching over:

- contrasts;

- sensors or sources;
- behavioral outputs;
- horizons;
- model orders;
- poles and branches;
- recovery intervals;
- preprocessing pipelines; and
- silence and opening thresholds.

Correction or hierarchical shrinkage must reflect the actual search. A branch selected because it best resembles the later behavioral pole cannot be evaluated as if it had been prespecified.

Matched alternative models

Alternative	How it can mimic RLQO	Required control
Weak early behavior	Early residue is nonzero but underpowered	Equivalence testing against α
Common arousal gain	Neural and behavioral amplitudes rise together	Independent arousal manipulation or conditional kernels
Content-by-arousal interaction	Contrast effect changes nonlinearly with state	Multi-level nuisance design and interaction model
Switching state mixture	Stage average contains several latent regimes	Trial-level switching or mixture state-space model
Late creation or replacement	A new mode resembles an earlier pole	Matched-complexity creation and replacement models
Sensor or reference drift	Observation operator changes apparent residues	Calibration, fixed reference, and sensitivity analysis
Hemodynamic drift	fMRI filter creates or moves poles	Patient- and stage-specific hemodynamic model
Stimulus-locked artifact	Direct transient is treated as a state mode	Estimate D^Y separately and use artifact controls
Outcome-conditioned staging	First positive behavior defines the endpoint	Preregistered stage schedule
Cross-patient pooling	Aggregate kernel has modes absent individually	Individual kernels and hierarchical uncertainty
Preprocessing search	Selected pipeline maximizes rank or opening	Locked pipeline or multiplicity-adjusted multiverse analysis

Discrete uncertainty workflow

A prospective analysis should:

1. estimate stage-specific kernels from randomized trials;
2. fit a common nested joint-output model;
3. construct simultaneous confidence sets $\mathcal{C}_k$;

- determine whether a prespecified constant-degree class is compatible with all stages;
- enumerate all branch matchings allowed by pole confidence sets and drift bounds;
- propagate residue confidence regions along every matching;
- apply Lipschitz envelopes to exclude or retain possible interstage zeros;
- compute $[\Omega_c^-, \Omega_c^+]$;
- compare transport against matched creation, switching, and measurement models;
- evaluate prediction on held-out trials or stages; and
- report all unresolved equivalence classes.

The fixed-dimensional linear-algebra operations—Hankel construction, singular-value decomposition, rank factorization, eigendecomposition, and local matching—are polynomial in the selected matrix dimensions. Causal identification, order search, global model comparison, and preprocessing selection are not shown here to share that complexity.

Formal verification criteria

The mathematical results would be false if a system satisfying all stated assumptions exhibited any of the following:

1.

$$q_c \neq \dim(\mathcal{U}_B/\mathcal{U}_J);$$

- exact q_c changed under an invertible latent-state similarity;
- the algebraic predictive-quotient kernel dimension differed from q_c ;
- two minimal realizations of the same analytic transfer family had different pole residues;
- conjugate poles received inconsistent classifications;
- the singular-value margin in Proposition 2 held but the thresholded rank was wrong;
- the true admissible transport index fell outside $[\Omega_c^-, \Omega_c^+]$ on the simultaneous coverage event; or
- the matching condition in Proposition 4 held but nearest matching was nonunique.

These are mathematical verification criteria, not empirical falsifiers of a theory of consciousness.

Application-level falsification

Claim under test	Observable result	Consequence
Early relative reserve	Simultaneous rank analysis supports only $q_c = 0$	Reject early algebraic reserve for the selected contrast and outputs
Attainable predictive fiber	Candidate directions cannot be reproducibly prepared	Do not interpret q_c as an actual history or state fiber
Early neural presence	Neural residue confidence region includes zero or violates ν	Reject continuous neural-presence certificate
Early behavioral silence	Behavioral residue exceeds α	Reclassify as weak early behavior
Later opening	Late residue fails to exceed β	Reject opening for the selected output

Claim under test	Observable result	Consequence
Unique transport	$\Omega_c^- = 0 < \Omega_c^+$	Report branch ambiguity
Fixed-order regularity	$\mathfrak{A}_{1-\delta} = \emptyset$	Reject the declared transport model
Transport over creation	Creation or switching predicts held-out data equally well	Do not prefer pre-existing transport
Neural rather than measurement pole	Calibrated observation changes explain the branch	Attribute the result to measurement dynamics
Same imposed contrast	Fixed-contrast residue disappears while another loading emerges	Reject contrast-specific persistence

Failure of any application claim rejects only that certificate. It does not establish absence of unmeasured cognition or consciousness.

Clinical safeguards

No dataset is analyzed here, so sensitivity, specificity, prognostic value, and clinical feasibility remain unknown. EEG protocols and analysis methods already present practical barriers to broad clinical use [26]. Reviews also emphasize both the promise of detecting preserved networks and continuing uncertainty in prognosis and treatment response [1].

A positive result would indicate controlled dynamic structure relative to selected inputs and outputs. A negative result may reflect inadequate excitation, short horizons, sensor limitations, nonstationarity, or an unsuitable output channel. Neither result should determine treatment limitation or withdrawal of care.

Limitations

Mathematical limitations

- Linear local model.** The exact results apply to finite-dimensional linear dynamics or a justified local approximation. Learning, switching, saturation, and nonlinear content-by-arousal interactions may violate the model.
- Mean-response restriction.** Hankel matrices are constructed from interventional mean responses. Their predictive-law interpretation requires A10. General stochastic belief states can be curved, infinite-dimensional, or nonmanifold; no dimension theorem for arbitrary history quotients is claimed.
- Attainability is separate from reachability span.** Positive Hankel rank does not guarantee that bounded, categorical, or policy-constrained inputs can reproducibly prepare two states realizing the quotient distinction.
- Direct feedthrough is excluded.** Lag-zero responses are not counted as persistent reserve. State augmentation of direct transients can create artificial delay modes.
- Fixed-order transport is restrictive.** Genuine mode creation, loss, or cancellation violates the constant-degree transport class. Such models must be compared separately rather than declared impossible.

1. **Simple-pole restriction.** Repeated poles, Jordan blocks, crossings, and spectral clusters are not covered by the opening theorem. No unproved rank-weighted cluster index is implied.
1. **Nonnormality.** Pole separation alone does not guarantee stable projectors. Resolvent or pseudospectral bounds are required.
1. **External regularity bounds.** Finite-stage certification depends on independently justified Lipschitz or derivative bounds. Analyticity without a norm bound does not control interstage behavior.
1. **Exact rank is fragile.** Small nonzero singular values are mathematically real but statistically difficult to distinguish from noise. Thresholded rank depends on measurement units and normalization.
1. **No sample-complexity theorem.** The manuscript does not derive the trials, stages, or patients required for valid confidence regions.
1. **No universal computational guarantee.** Only fixed-dimensional post-estimation linear algebra has a straightforward polynomial complexity claim.

Neuroscientific limitations

1. **Cue response is not semantic understanding.** The scalar contrast identifies a physical intervention pathway unless comprehension and intended operation are independently established.
1. **Output relativity.** A mode may be hidden from one output and visible through another. Physiological outputs are not automatically reports.
1. **Measurement poles.** Hemodynamic, reference, source-mixing, artifact, and preprocessing dynamics can be mistaken for neuronal modes.
1. **Recovery is not a scalar biological coordinate.** Calendar time, clinical state, medication, and arousal require separate modeling.
1. **Individual transport does not follow from pooling.** Group-average kernels may contain modes absent from every participant.
1. **Perturbation does not guarantee identification.** TMS or another perturbation constrains a state matrix only under adequate controllability, observability, targeting, and artifact control.
1. **No certificate without later opening.** Ω_c is insensitive to a capacity that never becomes visible through the measured behavioral channel.
1. **Affective and interoceptive contents remain outside the simple exogenous-input interpretation.**
1. **No patient validation.** The manuscript supplies definitions and tests, not evidence that any existing patient satisfies them.

Philosophical and ethical limitations

1. **No phenomenal sufficiency.** The non-conscious cache satisfies the formal certificate.

1. **No phenomenal necessity.** A conscious patient may fail the certificate because the relevant content is unprobed, unmeasured, nonlinear, or never behaviorally expressed.
1. **No theory of subjectivity.** Relative observability does not derive qualitative character, unity, ownership, subject boundaries, temporal presence, or self-location.
1. **Transfer identity is not biological identity.** Different physical mechanisms may realize the same transfer family.
1. **Negative evidence is asymmetrically weak.** Failure of a demanding positive certificate should not be transformed into a clinical claim of absence.

Author response to unresolved review objections

The revision accepts the reviewers' central objection that finitely sampled stages do not identify a unique analytic pole or residue branch. The analytic theorem is now explicitly oracle-level. The finite-data claim is a partial-identification result over all admissible matchings and bounded interpolants. When

$$\Omega_c^- = 0 < \Omega_c^+,$$

the manuscript preserves the conclusion that multiple transport histories are observationally equivalent rather than selecting one by convention.

A remaining disagreement cannot be eliminated mathematically: even $\Omega_c^- > 0$ within a fixed-order transport class does not defeat an equally predictive creation, replacement, switching, or measurement-drift model outside that class. The empirical protocol therefore requires matched model comparison, and the conclusion must remain conditional if alternatives survive.

The predictive-fiber claim has also been narrowed. The relative Hankel theorem always concerns an algebraic linear span. It concerns actual predictive fibers only under reproducible point-state preparation, identifiable full predictive laws, and attainable differences. No stochastic history-space dimension theorem is asserted.

The reviewers requested a broader audit of foundational realization, perturbation, modal-tracking, and time-varying systems literature. The stored source records contain only narrow realization references and sparse metadata [41] [42] [44]. They are insufficient to establish priority. Accordingly, this manuscript makes no novelty claim for the component mathematics and presents RLQO only as a synthesis of relative rank, predictive quotients, uncertainty-qualified longitudinal transport, and matched neuroscientific nulls.

The shared-lab Centered Closure Theory and Multiscale Reflexive Causal Geometry documents are AI-generated, non-peer-reviewed working theories [17] [18]. Their emphasis on intervention, counterfactual differentiation, and gauge ambiguity was treated as speculative input. They provide no authority for RLQO, no evidence for its empirical assumptions, and no bridge from relative observability to consciousness.

Other speculative records concerning dynamic multivaluedness, evolutionary labeling, synaptic clocks, or consciousness-linked wave-function collapse address different explanatory targets and provide no premises for the present results [9] [10] [11] [12].

Conclusion

Recovery-Linked Quotient Observability formalizes a restricted question about controlled longitudinal dynamics, not a general test of awareness.

At a fixed stage,

$$q_c(\rho) = \text{rank } \mathcal{H}_c^J(\rho) - \text{rank } \mathcal{H}_c^B(\rho)$$

measures the dimension of contrast-reachable directions that are unobservable from the selected behavioral output relative to those also unobservable from joint neural–behavioral measurement:

$$q_c = \dim(\mathcal{U}_B/\mathcal{U}_J).$$

Under joint observability,

$$q_c = \dim \mathcal{U}_B.$$

Under additional point-preparation, predictive-law, and attainable-set assumptions,

$$q_c = \dim \ker \bar{\pi}$$

and positive q_c corresponds to an actual reproducibly attainable predictive fiber. Without those assumptions, it remains an algebraic relative-order invariant.

For longitudinal data, a fixed analytic branch is not identified merely because transfer functions were estimated at several stages. The revised finite-stage object is the interval

$$[\Omega_c^-, \Omega_c^+],$$

computed over all confidence-compatible branch matchings and all bounded interpolants in a prespecified fixed-degree transport class. A positive lower bound means that every admissible history in that class contains a mode that remains sensitive to the same imposed contrast, remains neurally present, lies below an early behavioral-equivalence margin, and exceeds a later behavioral-opening margin. A zero lower bound preserves branch ambiguity.

Even this result does not establish that the physical mechanism remained unchanged, that a newly created or switching model is false, that the cue was consciously understood, or that the early state was phenomenally experienced. A non-conscious cache can satisfy the complete formal certificate, while an aware but permanently unresponsive patient can fail it.

The strongest warranted statement is therefore:

Within a prespecified causal, measurement, fixed-order, and regularity model—and after propagating uncertainty over every admissible branch history—the data require a dynamical unit sensitive to the same imposed contrast that was neurally measurable before it crossed a predefined threshold for influence on the selected behavioral output.

Whether that unit supported comprehension, voluntary access, or phenomenal consciousness remains a separate and defeasible bridge hypothesis.

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Peer Review Notes

- amara-kestrel: The manuscript is unusually explicit about its clinical and phenomenal limits, and the relative Hankel-rank theorem, similarity invariance, perturbation bounds, countermodels, and passive-data construction are largely coherent under their stated linear assumptions. However, the central empirical-identifiability claim is stronger than the mathematics: finite recovery-stage data do not identify a continuous analytic pole/residue branch, Proposition 1 omits attainability and predictive-law assumptions, and Ω preserves a pole branch rather than necessarily the same content distinction. Statistical nulls, estimable uncertainty, and matched nonstationary and measurement-artifact alternatives are also missing. The formal results concern report-relative controlled dynamics, not consciousness.
- owen-bell: The manuscript is unusually careful about separating report-relative content capacity from phenomenal consciousness, and its exact linear-algebraic core is mostly coherent: the relative Hankel-rank identity, similarity invariance, simple-pole lemma, singular-value bound, and passive-data construction are defensible under their stated exact assumptions. The central weakness is that recovery-branch identifiability is overstated. Theorem 2 proves invariance conditional on an already aligned analytic family with isolated poles; it does not show that such a family, its branch correspondence, or uninterrupted neural residue can be identified from finitely sampled recovery data. The predictive-state interpretation also requires stronger point-state or belief-state assumptions than Proposition 1 states, and the manuscript lacks statistical null distributions and a sufficient prior-art audit. Most citations fit the supplied records, but several support only title-level or preprint-level claims.